

MULTIMEDIA



UNIVERSITY

STUDENT ID NO

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MULTIMEDIA UNIVERSITY

FINAL EXAMINATION

TRIMESTER 2, 2016/2017

TCN 2141 – COMPUTER NETWORKS
(All Sections / Groups)

10 MARCH 2017
9.00 a.m. – 11.00 a.m.
(2 Hours)

INSTRUCTIONS TO STUDENTS

1. This Question paper consists of 5 printed pages including cover page with 6 questions only.
2. Attempt ALL questions. Marks and the distribution of marks for each question is given.
3. Please write all your answer in the Answer Booklet provided.

Question 1 [10 Marks]

- (a) Identify **FOUR** possible services that can be offered by a link-layer protocol. [2 Marks]
- (b) What is the purpose of ARP? Describe how ARP works. [4 Marks]
- (c) Describe the following controlled access mechanisms in a broadcast environment. [4 Marks]
- Polling
 - Token Passing

Question 2 [10 Marks]

- (a) Briefly describe Dynamic Host Configuration Protocol (DHCP) and identify the messages that are exchanged between the client and server in the DHCP operation as shown in Figure 1.

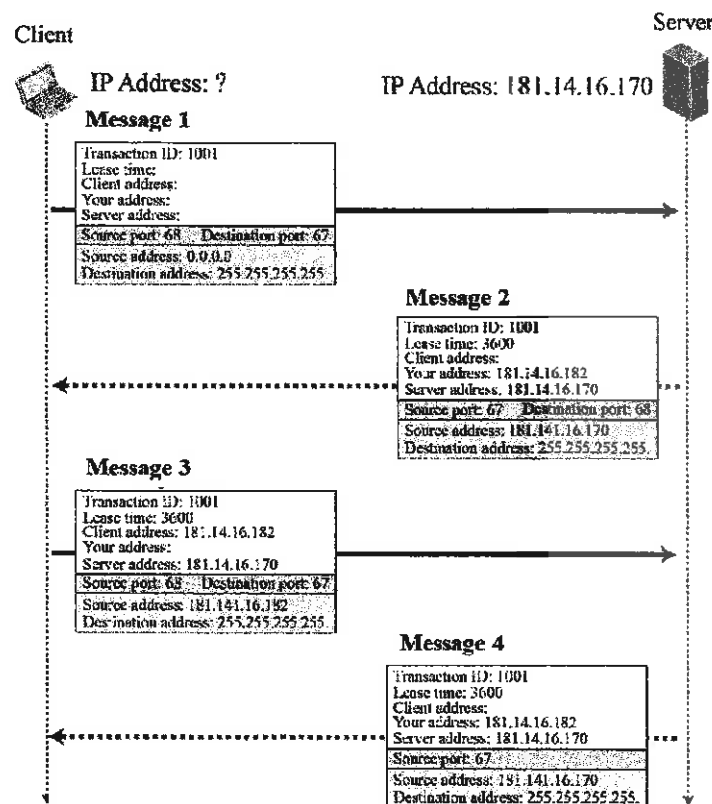


Figure 1

[4 Marks]

Continued ...

- (b) Explain the fundamental difference between datagram and virtual-circuit approach.

[4 Marks]

- (c) Discuss the difference between unicast routing and multicast routing.

[2 Marks]

Question 3 [10 Marks]

- (a) The Transmission Control Protocol (TCP) uses a handshake process to establish a connection between client and server. With the aid of a diagram, illustrate the TCP connection establishment process.

[4 Marks]

- (b) Describe the difference between transport mode and tunnel mode of IP Security (IPSec).

[2 Marks]

- (c) Describe the functions of the two File Transfer Protocol (FTP) connections.

[2 Marks]

- (d) Briefly explain what is Domain Name System (DNS).

[2 Marks]

Question 4 [10 Marks]

- (a) Discuss the concept of redundancy in error detection.

[3 Marks]

- (b) Cyclic Redundancy Check (CRC) is one of the well-known redundancy-based error detection techniques. Identify the other TWO types of redundancy checks used in data communications.

[2 Marks]

- (c) Consider a generator polynomial in a Cyclic Redundancy Check (CRC) error detecting scheme is $G(x) = x^3 + x^2 + 1$. Using polynomial division, determine the codeword for the message bits $m = 1011$.

[5 Marks]

Continued ...

Question 5 [10 Marks]

- (a) Identify **TWO** practical applications of multicasting.

[2 Marks]

- (b) Run Dijkstra's algorithm on the network as shown in Figure 2 to determine the routing table for SEL. Please show all intermediate steps.

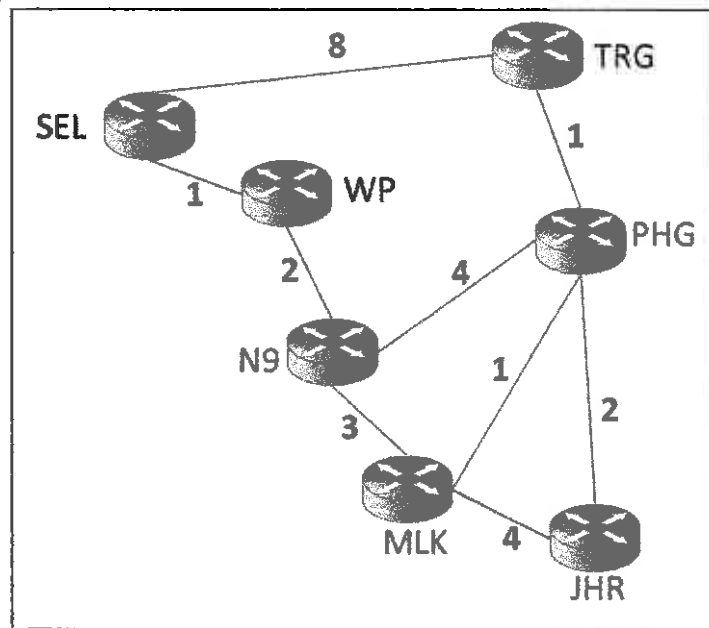


Figure 2

[4 Marks]

- (c)

Host IP Address	:	192.168.10.158
Network Mask	:	255.255.255.0
Subnet Mask	:	255.255.255.240

In a networking project, the above information is provided. Determine the following.

- Subnet address of the host.
- Broadcast address of the host.
- Usable IP addresses for this subnet

[1 + 1 + 2 = 4 Marks]

Continued ...

Question 6 [10 Marks]

- (a) Identify and explain the flow control protocol that is used in Figure 3.

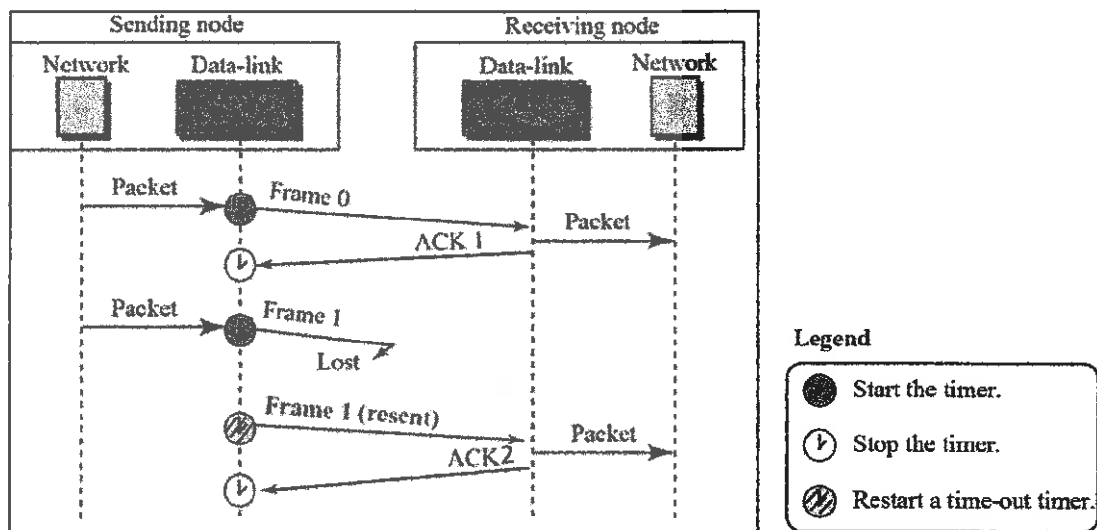


Figure 3

[3 Marks]

- (b) Describe on how IP fragmentation may impact in terms of analysing application traffic.

[2 Marks]

- (c) By referring to Figure 4, consider a 4020-octet datagram is to be transmitted from Host A to Host B and needs to be fragmented because it pass through an Ethernet with a maximum payload of 1500-octet. Suppose the original datagram is stamped with the identification number 777. Show the Identification, Total Length, Don't Fragment, More Flag, and Fragment Offset values in each of the resulting fragments.

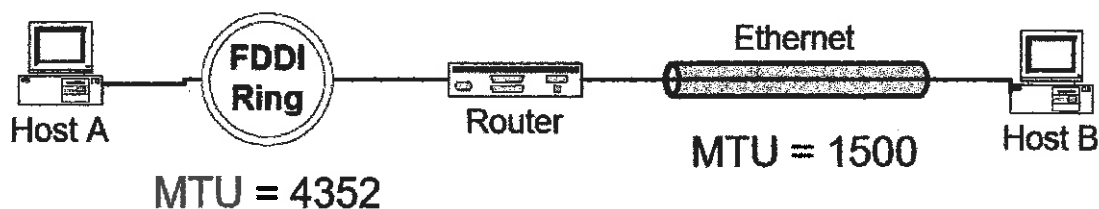


Figure 4

[5 Marks]

End of Paper